EVERY DIFFERENTIAL GRADED MODULE ADMITS A PROPERTY (P) RESOLUTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the inductive resolution through lemma-good-quotient, the total module with its explicit grading and differential, the interleaved filtration F_{2i} and F_{2i+1} with sums of shifts of A as quotients, the admissibility of the inclusions through lemma-target-graded-projective, and the quasi-isomorphism of the augmentation are all retained verbatim.

Editorial note. The label lemma-resolve names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Good quotient covers of differential graded modules. Same chapter, Lemma lemma-good-quotient. Supplies, at each induction step, the surjection P_i to M_i whose kernel continues the resolution and whose two-term structure produces the graded pieces.
- Admissibility of inclusions between sums of shifts and the resolution spectral sequence. Same chapter, Lemma lemma-target-graded-projective; Homology, Lemma homology-lemma-good-resolution-gives-qis. Shows the filtration inclusions are admissible monomorphisms, and turns the constructed resolution into the quasi-isomorphism P to M.

Source reference. Original lemma 09IZ: sources/stacks-chow/tags.txt, source lines 12057–12057. Prerequisite pointer invoked at proof line(s) 2565; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09KN: sources/stacks-dga/pinned-original-source.tex, source lines 2490–2490. Prerequisite pointer invoked at proof line(s) 2534; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09K0: sources/stacks-dga/pinned-original-source.tex, source lines 2106–2106. Prerequisite pointer invoked at proof line(s) 2561; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:dga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let (A, d) be a differential graded algebra. Let M be a differential graded A-module. There exists a homomorphism $P \to M$ of differential graded A-modules such that*

(1) *$P \to M$ is a quasi-isomorphism, and*
(2) *P has property (P).*

Proof. Set $M = M_0$. We inductively choose short exact sequences

$$0 \to M_{i+1} \to P_i \to M_i \to 0$$

where the maps $P_i \to M_i$ are chosen as in Lemma 09KN. This gives a "resolution"

$$\ldots \to P_2 \xrightarrow{f_2} P_1 \xrightarrow{f_1} P_0 \to M \to 0$$

Then we set

$$P = \bigoplus\nolimits_{i \geq 0} P_i$$

as an A-module with grading given by $P^n = \bigoplus_{a+b=n} P^b_{-a}$ and differential (as in the construction of the total complex associated to a double complex) by

$$\mathrm{d}_P(x) = f_{-a}(x) + (-1)^a \mathrm{d}_{P_{-a}}(x)$$

for $x \in P^b_{-a}$. With these conventions P is indeed a differential graded A-module. Recalling that each P_i has a two step filtration $0 \to P'_i \to P_i \to P''_i \to 0$ we set

$$F_{2i}P = \bigoplus\nolimits_{i \geq j \geq 0} P_j \subset \bigoplus\nolimits_{i \geq 0} P_i = P$$

and we add P'_{i+1} to $F_{2i}P$ to get F_{2i+1}. These are differential graded submodules and the successive quotients are direct sums of shifts of A. By Lemma 09K0 we see that the inclusions $F_iP \to F_{i+1}P$ are admissible monomorphisms. Finally, we have to show that the map $P \to M$ (given by the augmentation $P_0 \to M$) is a quasi-isomorphism. This follows from Homology, Lemma 09IZ. □